Auditing Generated KPI Documentation Against Source Code: A Study of 28 KPI Pages

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ABSTRACT

BAJ: Given the brevity, avoid the bolded headings; instead, write a sentences which naturally lead from one topic to the next. MES: the bold was deleted Software calculates key performance indicators (KPIs), but the documentation explaining those calculations may not accurately describe what the code does. Even when documentation links to the correct source code, its explanations may contain errors or omit conditions that affect the calculated result.

We evaluated this problem using LEAP (Literate Programming for Automated KPI Extraction), a research software tool that extracts KPI information from source code and publishes source-linked documentation in an Interactive Bluebook. For example, one generated KPI page presented a production-yield formula but omitted a condition under which the software returns zero.

We audited 28 generated KPI documentation pages using deterministic checking rules and assessments from three independent reviewers. The evaluation tested four candidate dimensions of documentation accuracy. Two received reproducible reviewer support: correct descriptions of source-code elements (Fleiss' $\kappa = 0.407$) and disclosure of conditions affecting calculations ($\kappa = 0.553$).

The automated source-role checker detected only 2 of 16 errors identified by reviewer majority (12.5% recall). For condition disclosure, checker precision differed between two assessment criteria: 40% when source visibility counted as disclosure and 100% under the reader-facing criterion used by the checker. The latter result reflected agreement with that criterion, not independent proof of perfect detection.

The study demonstrated that linking documentation to source code does not guarantee an accurate explanation. It also identified specific documentation errors and limitations of the automated rules used to detect them.

1. Introduction

BAJ: This paper needs a proper introduction: begin by stating the problem, then why this approach solves the problem. The material here comes after these paragraphs. Define KPIs, why these are important, how they're not traditionally linked to code, etc.

BAJ: [Tying to the need for LEAP...] Therefore,

MES: Key performance indicators (KPIs) are quantifiable measures used to evaluate progress toward strategic and operational objectives. Their meaning connects a business definition of what should be measured with the calculation that produces the reported value. In software-based reporting, this connection includes the inputs, formula, and conditions implemented in source code. Understanding a KPI therefore requires access to both its business interpretation and the computational decisions that determine its value (Maté et al., 2017; Roldán-García et al., 2021).

MES: In the manual workflow motivating this study, the business definition, software, and documentation are maintained separately. A business analyst specifies the indicator, a developer implements its calculation, and a separate reference document explains it to business readers. When a reported value is questioned or the code changes, the team must compare that document with the implementation to check which inputs, formula, and exceptional cases apply. The code produces the number; the document is meant to explain its meaning. Research on documentation maintenance and code–comment consistency shows why that connection must be maintained as software evolves (Aghajani et al., 2019; Wen et al., 2019; Xu et al., 2024).

MES: LEAP (Literate Programming for Automated KPI Extraction) is a research software tool that scans source files for potential KPI calculations and generates documentation pages linked to the code. Its collection of pages is called the Interactive Bluebook; each page is called a dossier. A dossier can show the reported formula, source-code location, descriptions of the code, safeguards, and review status. The links help people find the source, but they do not guarantee that the generated descriptions are correct.

We checked whether 28 documentation pages generated by LEAP accurately explained their linked calculations. We compared the published descriptions with the source code, traced discrepancies to the generator, and compared automated checker decisions with assessments from three independent reviewers. The audit focused on the information a reader needed to understand how the software calculated each KPI.

For a concrete example, consider Ethylene Production Yield. The ordinary formula divides the tons of ethylene produced by the tons of feedstock supplied, then multiplies by 100. However, the archived source returns zero whenever the feedstock input is zero or negative. LEAP publishes the ordinary ratio and retains a link to the code, but the

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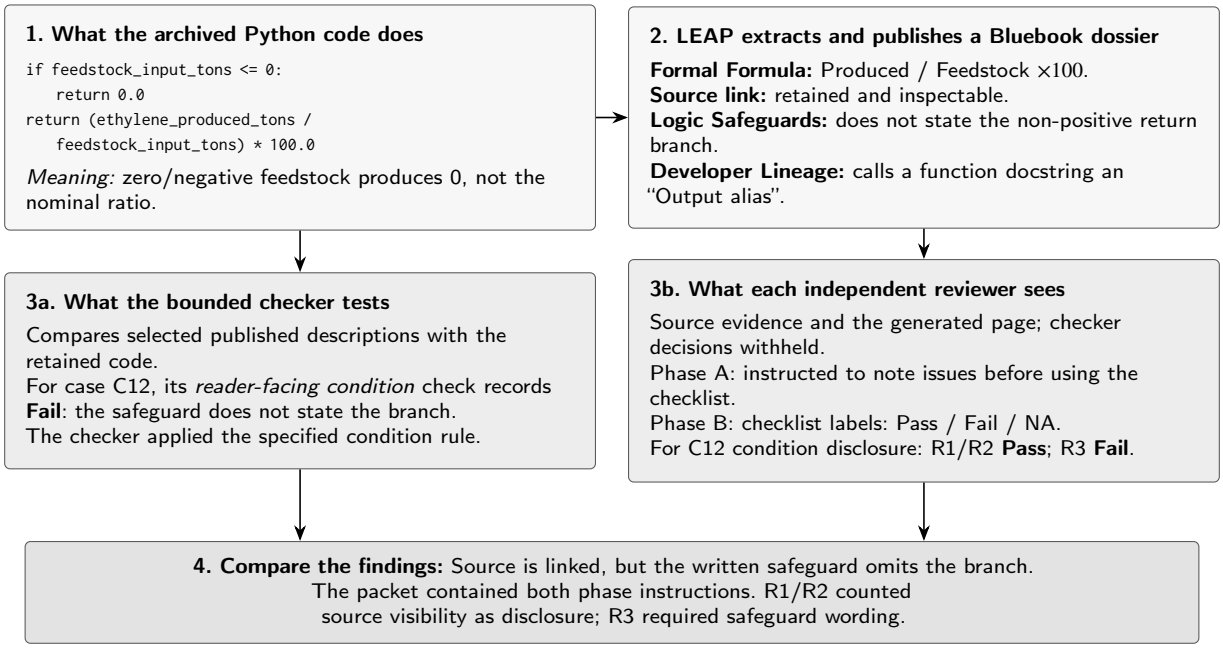


Figure 1: Schematic of the archived Ethylene Production Yield source, generated page, and recorded audit decisions. The checker assessed the condition using its reader-facing rule; reviewers assessed the supplied source and page independently of the checker outputs. Their condition labels differed over what counted as disclosure.

Table 1
Illustrative outputs of the archived Yield function for positive, zero, and negative feedstock inputs.

Ethylene produced (tons)	Feedstock input (tons)	Returned yield (%)
20	100	20
20	0	0
20	−10	0

Bluebook's Logic Safeguards text does not tell the reader about that branch. Someone reading only the documented ratio cannot see what the software will report for those inputs. Figure 1 follows this example through LEAP, the audit, and the independent reviewers. Table 1 gives illustrative outputs for the same function.

The example separates source access from explanation accuracy: the page links to the code yet leaves out a rule needed to interpret the returned value. The Gross Margin example in Section 5.2 examines a different problem, a mismatch between the stated business formula and the executable expression. We audited the documentation of both calculations.

We use *explanation fidelity* to mean whether the generated text correctly describes its linked source. We assessed four questions: whether descriptions identified source elements correctly, whether they stated conditions affecting the calculation, whether marked conflicts remained visible for review, and whether certification wording matched the recorded review status. We then compared the checker with the reviewers and inspected the generator code to explain repeated errors.

Prior work establishes the underlying problems. Code–text consistency research measures divergence between source and documentation (Wen et al., 2019; Xu et al., 2024; Eghbali et al., 2026), and API-documentation research detects missing or inconsistent parameter, constraint, and exception directives (Zhou et al., 2017). BAJ: What about traditional literate programming approaches, or documentation generators such as Doxygen, Javadoc, Rustdoc, etc.? These do an excellent job of code–text consistency.

MES: Literate programming provides an established foundation for connecting implementation with human-readable explanation. Knuth's WEB approach interleaves procedural code and narrative so that readers can follow a program's rationale and structure alongside its executable components (Knuth, 1984).

MES: Documentation generators provide further mechanisms for publishing source-connected explanations. Doxygen incorporates structured documentation comments, Javadoc associates documentation with Java declarations, and rustdoc compiles and runs suitable code examples as documentation tests (Doxygen, n.d.; Oracle, n.d.; The Rust Project Developers, n.d.).

MES: LEAP brings the business formula, source code, explanations, safeguards, and review status together on one KPI page. We examined whether the descriptions on that page accurately communicated the linked implementation.

The contribution is an audit procedure that connects an inaccurate KPI explanation to the source evidence and the generator code responsible for it. The reviewer comparison separately measured which assessment criteria reviewers

applied consistently and which of their findings the automated rules detected.

Reviewer judgments supported source-role accuracy and condition disclosure as the two retained criteria. The source-role checker detected 12.5% of reviewer-majority failures. Condition precision was 40% or 100%, depending on which disclosure criterion supplied the comparison labels. The disagreement and review-authority results led us to narrow the former to a LEAP-specific routing check and reject the latter as a reliable assessment criterion.

These findings yielded three contributions:

1. an audit that compares generated KPI descriptions with their source code and traces discrepancies to the code that generated the documentation;

2. an independent assessment of four proposed criteria on 28 pages, retaining source-role accuracy and condition disclosure, narrowing disagreement to explicit-conflict routing, and rejecting review-authority consistency; and

3. reproducible comparisons showing which discrepancies remained after source links were verified, which reviewer findings the checker missed, and how its measured precision changed with the definition of disclosure.

2. Related Work and Basis of the Assessment Criteria

2.1. Accuracy of Source Descriptions and Conditions

Software documentation remains difficult to create and maintain (Aghajani et al., 2019). Traceability identifies relationships between documentation and software artifacts (Mäder and Gotel, 2012); code-comment inconsistency studies show that linked text can still diverge from code semantics (Wen et al., 2019; Xu et al., 2024). Eghbali et al. formalize code-documentation equivalence around accurate and complete description of program behavior (Eghbali et al., 2026). Zhou et al. compare API code with parameter-constraint and exception directives to detect documentation defects (Zhou et al., 2017). BAJ: Need another sentence– "This research demonstrates the ongoing problem of failed code–documentation equivalence. The LEAP system addresses this problems by..." MES: These studies identify errors in explanations of code and its behavior. LEAP places the source beside generated formulas, explanations, and safeguards; we compared those fields with the retained source. This comparison applied the two requirements motivated by the cited work: describe source elements correctly and state the conditions that qualify their behavior.

We applied these consistency requirements to generated KPI pages. BAJ: Then why reference these as related works? MES: Each audited page already contained a source link. Holding that source evidence fixed let us ask a separate question: did the published explanation correctly describe the role of the source element and the conditions affecting its behavior?

2.2. Disagreement, evidence roles, and authority

Nuseibeh et al. show that inconsistency can be retained and managed rather than silently eliminated (Nuseibeh et al., 2001). SEPIO BAJ: A bit confusing – SEPIO is a new acronym not defined yet, but appears in the referenced paper. State more explicitly that SEPIO is the name of the system in this paper. The same goes for the following acronyms. MES: (Scientific Evidence and Provenance Information Ontology), the evidence model introduced by Brush et al., links claims to supporting or refuting evidence (Brush et al., 2016); ECO MES: (Evidence and Conclusion Ontology) distinguishes types of evidence and how assertions were made (Giglio et al., 2019). These works motivated two further questions: does a page preserve an explicitly marked conflict, and does its certification wording agree with the recorded review status? BAJ: Tie this back to LEAP. How does LEAP do this? Does it merely follow this existing work, or does it provide a new contribution? MES: LEAP retains the source representations, conflict markers, and review status on the same page. We checked whether marked conflicts remained routed for review and whether certification wording agreed with that status. We assessed reviewer agreement on both proposed criteria separately from the behavior of the corresponding automated rules.

PROV MES: (the World Wide Web Consortium's family of provenance specifications) describes evidence origins through entities, activities, and responsible agents, with rules for valid provenance records (Missier et al., 2013; World Wide Web Consortium, 2013). OneProvenance reconstructs dynamic query lineage from database events (Psallidas et al., 2023). ESDPKI MES: (Enterprise Spatial Data Provenance Knowledge Infrastructure) combines provenance records with validation, publication, governance, and access controls (Sadiq et al., 2026). These systems distinguish where evidence came from, how it was checked, and how it was managed. Our audit asked whether the KPI page communicated its source and review status accurately. BAJ: Again, how does this affect LEAP? How do you distinguish your contribution from earlier work? Same for following paragraphs. MES: For a LEAP reader, a source link identifies the implementation behind a KPI, while the review status states how the page is classified. We examined the descriptions accompanying those records. This separates the existence of source and status records from the accuracy of what the page tells the reader.

2.3. Source-bound software knowledge

ECK MES: (Executable Code Knowledge) organizes executable knowledge around source evidence, validation state, freshness, and checks of derived representations (Gao, 2026). Reversa preserves traceability, confidence, and explicit validation gaps in reconstructed specifications (de Macedo and da Costa, 2026). These approaches motivated checking what happens when evidence is presented in another form. Our checklist examined the generated descriptions, calculation conditions, explicit conflicts, and review-status wording on KPI pages.

Table 2
Literature basis and implemented checks for the four proposed assessment criteria.

Assessment criterion	Established basis	Implemented LEAP check
Source-role accuracy	Code–text consistency and documentation equivalence (Wen et al., 2019; Xu et al., 2024; Eghbali et al., 2026)	Flag a published explanation that assigns a source element the wrong syntactic/semantic role.
Condition disclosure	Completeness of behavioral/API directives (Eghbali et al., 2026; Zhou et al., 2017)	For explicit zero/non-positive denominator guards, preserve the guarded variable and condition semantics.
Representation disagreement	Inconsistency management and conflicting evidence (Nuseibeh et al., 2001; Brush et al., 2016)	For source evidence carrying an explicit conflict marker, preserve the conflict in review routing.
Review-authority consistency	Evidence/assertion roles, provenance validity, and validation-governance separation (Giglio et al., 2019; World Wide Web Consortium, 2013; Sadiq et al., 2026)	On unresolved dossiers, flag bounded certification-oriented wording.

BAJ: Summarize: while previous research shows the need for... LEAP therefore provides these needed enhancements, by... as detailed in the following section. MES: Together, these studies motivate checking both the connection to source code and the accuracy of its explanation. LEAP provides the page on which code, formula, safeguards, and review information can be inspected together. We evaluated the four proposed assessment criteria using reviewer agreement and measured how the implemented checker compared with the recorded judgments.

KPI semantics provide the domain context. Semantic KPI work formalizes indicator definitions and derivations (Maté et al., 2017; Roldán-García et al., 2021; Diamantini et al., 2025), and AFFECTK MES: (the KPI-evolution framework introduced by Dominguez et al.) makes traceability and consistency explicit during KPI evolution (Domínguez et al., 2025). We evaluated the four proposed criteria on pages that combined KPI formulas, source evidence, and review status. Reviewer judgments informed the choice of retained criteria; checker comparisons identified missed discrepancies. BAJ: You need to show how LEAP improves on earlier designs, or offers new abilities. RQs about software behavior are less relevant here. What's the core contribution LEAP provides in this context? MES: The contribution is a completed audit of generated KPI explanations: we located discrepancies against linked code, traced their causes, and compared automated checks with independent reviewer judgments. This distinguishes the usefulness of an assessment criterion from the performance of the rule used to implement it.

3. Research Questions and Method

The completed study addressed the following research questions:

RQ1 Which MES: discrepancies between source evidence and published KPI explanations are exposed in the full 28-dossier LEAP output, and which generator mechanisms explain the repeated manifestations?

RQ2 Which MES: discrepancies in published KPI knowledge persist or vary across an earlier project state and alternative input configurations?

RQ3 Which candidate explanation-fidelity dimensions achieve reproducible independent reviewer agreement?

RQ4 What checker failure modes and criterion sensitivities appear when the released rules are compared with independently produced reviewer labels?

3.1. Claims and falsification conditions

Table 3 links each main claim to its evidence and states what observation would overturn it.

3.2. Case-study system and publication pipeline

There were four distinct roles in the evaluation. LEAP generated the source-linked KPI pages. The checker applied predetermined rules to the source and published fields. The reviewers inspected the cases and recorded judgments. We compared those judgments with checker outputs and inspected the generator code to trace repeated discrepancies. Figure 2 separates page generation from the checker, reviewer, and comparison steps. LEAP generates documentation from KPI source evidence. Its BAJ: [seems extraneous; do you means that this is frozen for publication purposes?] frozen research artifact MES: implementation scans source files for KPI candidates, prepares formula and business-context fields, attaches source links and review information, and generates an HTML page for each selected KPI. The page contains source context, a displayed formula, descriptions of source elements, safeguards, ownership information, conflict markers, and a review status. The source-element descriptions appear in its Developer Lineage field. We audited those published fields and traced repeated errors to the BAJ: [Again, distracting from the goal of the sentence] frozen generator generator implementation. MES: the repeated frozen qualifier was deleted

Table 3
Major claims, supporting evidence, and falsification conditions.

Claim	Supporting evidence	Falsification condition
Within the LEAP case study, source linkage can coexist with publication discrepancies.	Source-linked Yield and Gross Margin dossiers retain inspectable source evidence while direct comparison identifies role and condition discrepancies.	The claim fails if the retained evidence link is absent or direct source comparison removes the reported discrepancy.
Three parts of the LEAP generator produced the repeated documentation errors.	Source inspection traced source-role descriptions, safeguards, and certification wording to three shared parts of the generator.	The claim fails if rerunning the released audit changes the manifestations or if repeated manifestations map to different generator mechanisms.
Independent review differentiates the candidate dimensions.	Three reviewers assessed 112 page–criterion combinations; Fleiss' κ ranges from 0.553 to BAJ: Why does this render as *0.285? -0.285 MES: the minus sign was corrected, and restricting the disagreement analysis to cases judged applicable by all three reviewers changed the result.	The claim fails if the preserved reviewer forms and deterministic aggregation reproduce uniform agreement across the four candidates.
The released source-role rule fails coverage, while condition-rule performance is criterion-sensitive.	Source-role recall is 12.5% against reviewer majority. On the 10 checker-eligible condition cases, precision is 40% under the source-visibility criterion and 100% under the reader-facing criterion; recall is 100% under both.	The claim fails if deterministic recomputation removes the source-role confusion matrix or the dual-criterion condition results.

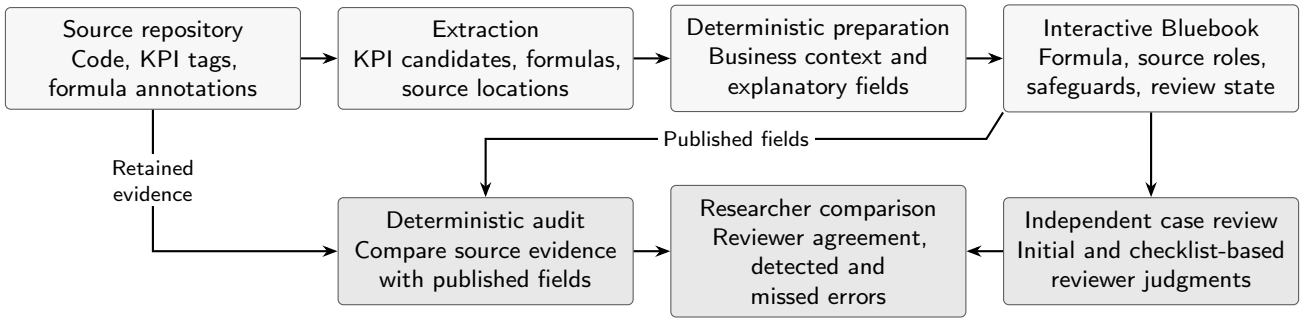


Figure 2: The evaluated LEAP workflow (upper row) and the audit and reviewer comparison (lower row). LEAP generated the pages; the checker and reviewers assessed them separately; the researchers compared their decisions. The archived generation log records deterministic detail preparation for all 28 pages.

3.3. Case-study artifacts and within-system design

MES: Here, frozen denotes the fixed evaluation snapshot of the generator, inputs, and published pages used to reproduce the reported comparisons. The primary case-study artifact contains all 28 rendered dossier HTML pages produced by the frozen LEAP sample-project build. A public artifact copy at commit `5e697b5b` (full identity in the supplement) matches all 28 submitted pages byte-for-byte by SHA-256. The build contains 9 *Validated* and 19 *Needs Review* dossiers. These recorded states support the concrete LEAP certification-string analysis and also provide evidence for evaluating the broader review-authority candidate independently.

We compared the earlier archived output of 18 pages, including 6 *Validated* and 12 *Needs Review* pages, to determine whether the same errors recurred. Two further input configurations used the same generator: 38 scenario-fixture files produced 29 pages, and 8 challenge files produced 8. These comparisons tested whether the findings changed with the input configuration. We treated each generated page as a case and assessed its source descriptions, calculation conditions, conflict markers, and review-status wording.

External adapter check. We also checked that the source-role and condition adapters executed on four official FastAPI tutorial sources and their generated OpenAPI 3.1 descriptions at fixed upstream identities (FastAPI Project, 2026; OpenAPI Initiative, 2026). These runs tested whether the adapters could read another documentation format. The independent reviewer evaluation used the LEAP pages.

3.4. Automated Checking Rules

BAJ: Why are these rules sufficient? What inputs cause them to produce incorrect results? The checker applied the following four rules. Their specified patterns determined what

each rule could flag; the reviewer comparison measured the discrepancies those patterns detected or missed. MES: The rules implemented specific checks motivated by the literature in Table 2. Prepared controls tested whether each rule returned its specified decision. Comparison with independent reviewers tested how many of their findings it detected. For source roles, it detected only 2 of 16 reviewer-majority failures.

1. **Source-role accuracy.** The checker flags a failure when a comment or docstring is described as executable code, or when a SQL `CAST` type is labeled as the output alias instead of the actual alias. The rule reads pairs consisting of a source excerpt and its generated Developer Lineage description.

2. **Executable-guard disclosure.** Eligibility covers explicit zero or non-positive denominator BAJ: Why is a negative denominator a problem? For zero checks, does it also include values that are close to zero? What about other places where the answer is problematic, such as negative values for a square root, negative values for an inverse log, invalid values for arc sin/cos/tan, etc.? guards expressed as `NULLIF(variable,0)` or an `if`-family comparison to zero. Preservation requires reader-facing safeguard text to communicate zero/non-positive handling and bind the guarded variable to the denominator displayed in the formal formula. A `<= 0` guard requires non-positive semantics; zero-only wording fails the rule. MES: A negative nonzero denominator is mathematically valid. The obligation here arises from the implemented branch: the Yield source returns zero when feedstock input is non-positive. Consequently, zero-only safeguard wording omits part of the implemented condition.

3. **Representation-disagreement candidate.** The released rule covers pages whose retained source explicitly carries a conflict, mismatch, or inversion marker and checks preservation through *Needs Review* plus a Review Queue. We separately measured reviewer agreement on the broader disagreement criterion.

4. **Review-authority candidate.** The released rule checks unresolved *Needs Review* dossiers for bounded certification-oriented wording such as “Certified metric dossier”, “certified calculation expression”, or “Print Certified Dossier”. Reviewer agreement determined whether this criterion was retained.

The scripts produced the full guard inventory and line-level source-role findings. Table 2 records the literature basis of the four criteria.

3.5. Evaluated checker and conformance tests

We used the R3 BAJ: does this mean rule 3? MES: (the third protocol revision) audit protocol and the R4 MES: (the fourth implementation revision) checker. The evaluated implementation compares published roles and names with their source counterparts, requires a nonempty displayed denominator, and tests safeguard wording for variable correspondence, handling semantics, and contradictions. The conformance tests below exercise these requirements through prespecified control outcomes.

We tested the checker against two sets of author-specified controls with expected labels stored separately from its output. Eighteen rule fixtures covered correct and incorrect source roles, missing or mismatched denominator information, omitted or contradictory safeguard text, conflict routing, certification wording, and applicability. Fourteen controls derived from the archived artifacts included five unchanged baselines, eight substitutions of extracted inputs, and one HTML mutation read again through the same page adapter. We compared each returned decision with its expected label. BAJ: Reviews will be suspicious here – the test found no failures for any of the rules? This would be more convincing if failures are in areas that humans will catch, etc. Matching the control oracle includes returning Fail when a control is deliberately defective. Across the 32 controls, the expected outcomes comprise 19 Fail, 11 Pass, and two N/A labels. These results establish conformance to the specified control cases. Independent comparison on the original dossiers reveals the coverage limitation: the source-role checker misses 14 of the 16 reviewer-majority failures.

3.6. Independent Review: Initial and Checklist-Based Phases

Three independent reviewers evaluated the same 28 pages, identified as C01–C28. The packets excluded checker outputs, manuscript result tables, the organizer mapping key, and other reviewers’ responses. In Phase A, reviewers were instructed to record issues using their own judgment before applying the four-criterion checklist in Phase B. The packet contained the instructions for both phases. Phase B recorded Pass, Fail, or NA with supporting evidence; NA meant that the criterion was judged inapplicable.

Each of the 28 pages was assessed against four criteria, producing 112 page–criterion combinations. Three ratings per combination produced 336 individual ratings. All 28 Phase-A case rows were returned by each reviewer. The original Phase-B responses contained 242 Pass/Fail ratings and 94 explicit NA ratings (28.0%). The 94 NA entries appeared as blanks in an intermediate analysis export: 40 of 84 condition ratings and 54 of 84 disagreement ratings. Restoring those entries to NA reproduced all 336 original labels. We also calculated agreement on the cases for which all three reviewers supplied Pass or Fail, indicating common applicability.

A majority decision required at least two matching ratings for a page–criterion combination. One condition case had a three-way split, leaving 27 resolved cases for that comparison. The other three criteria had 28 resolved cases each. We calculated average pair agreement, the fraction of matching labels across reviewer pairs, and Fleiss’ κ, an agreement measure adjusted for agreement expected from the label frequencies (Fleiss, 1971). We compared checker decisions with the reviewer-majority labels. Exact agreement

required the same Pass, Fail, or NA decision. For the binary comparison, Fail was positive and Pass/NA was non-fail.

A true positive (TP) was a checker flag supported by the majority reference; a false positive (FP) was an unsupported flag. A false negative (FN) was a majority-reference failure missed by the checker; a true negative (TN) was a non-fail case under both. Precision was the fraction of checker flags supported by the reference, and recall was the fraction of reference failures flagged by the checker. Exact / n in the results tables reports identical three-state decisions out of the resolved comparisons. The separate condition-criterion comparison used the same 10 checker-eligible cases under each disclosure definition.

The organizer guidance specified three reviewers, Fleiss' kappa, majority-agreement reporting, and a later adjudication step. The reviewer-packet README instead recommended two reviewers and Cohen's kappa, also followed by adjudication. The reported analysis used the three returned forms, Fleiss' kappa, and the stated majority rule as the comparison reference. We preserved the individual ratings and documented the protocol difference in the supplement.

Phase B followed Phase A on the same cases. We compared the majority issue sets and inspected the five cases flagged only in Phase B to identify which criterion accounted for the difference.

3.7. Verification and root-cause procedure

The primary audit script checked the page count, SHA-256 file identities, review-status counts, source-role descriptions, guard conditions, conflict routing, and certification wording. Primary, historical, alternative-input, and external scripts used the same released rule library for applicable checks. The primary scanner identified ten eligible guard cases. The release verifier checked the expected control labels and both conformance-test sets before accepting the corpus summaries.

After identifying documentation errors, we inspected the archived generator code to trace their causes. We linked repeated errors to a common cause when the same code explained their occurrence across pages. The supplement retains the generator excerpts and file hashes used in this inspection.

4. Results

4.1. Documentation Errors and Preserved Conflict Routing

The audit identified three causes of repeated documentation errors in the 28-page output: incorrect source-role descriptions, safeguards that lost the condition or variable checked by the code, and certification wording that appeared regardless of review status. All six pages with explicit conflict markers retained their review routing. Table 4 separates these generator causes from their occurrences on pages.

The alternative-input runs reproduced source-role and certification-wording findings. The 29-page scenario output had one source-role finding and certification wording on all 11 unresolved pages; its one eligible zero guard passed. The 8-page challenge output had one source-role finding and certification wording on all six unresolved pages; its one eligible zero guard also passed. Marked conflicts retained their review routing in both outputs. Guard results varied with the input configuration, while the other two findings recurred.

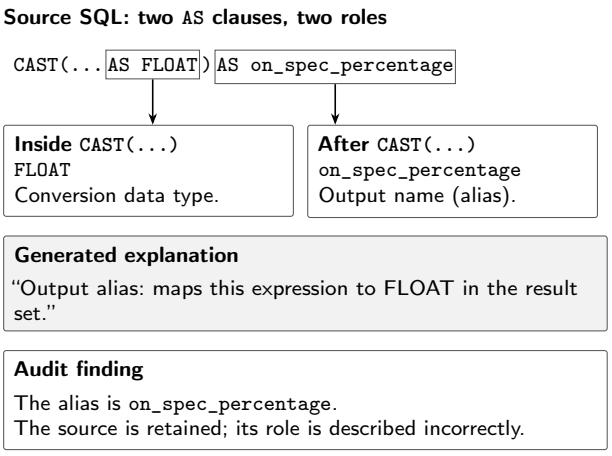


Figure 3: On-Spec Rate source-role error. The ellipsis omits arithmetic from the archived SQL statement. The two boxed AS clauses identify the conversion type and the output name; the generated Developer Lineage text confuses them.

4.2. Root-cause localization

Source inspection traced the repeated documentation errors to three parts of the generator. First, the developer-lineage classifier tests calculation/formula keywords before comment typing, leaves the ABAP * comment prefix outside its recognized comment prefixes, and applies a generic SQL AS pattern that can select a CAST type instead of the final alias. This classifier explains the source-role manifestations. Figure 3 illustrates this error on the On-Spec Rate page. The first AS specifies the conversion type, FLOAT; the second names the output, on_spec_percentage. The generator selected the type as the alias. The page retained the source, yet described the wrong element as the output name.

Second, safeguard generation recognizes NULLIF and selected equality-to-zero patterns and emits generic "zero denominator" language. In the affected pages, the text failed to identify which variable was guarded or to describe the non-positive-input branch. This mechanism explains the five current guard failures while correctly aligned NULLIF cases pass.

Third, certification-oriented strings are emitted independently of review state in generator prose and the page template. That mechanism explains the 19/19 certification-wording findings on unresolved pages. Explicit conflict signals follow a separate routing path to *Needs Review* and the Review Queue, producing the 6/6 preserved-conflict result.

4.3. Checker conformance-test results

The checker returned the expected result for all 18 rule fixtures and 14 artifact-derived controls: 19 Fail, 11 Pass, and

Table 4
Repeated documentation errors and conflict routing in the current and earlier archived LEAP outputs.

Class / mechanism	Current 28-dossier build	Earlier 18-dossier build	Interpretation
Lineage typing defect	4 dossiers; 12 manifestations	Same 4 dossiers; same 12 manifestations	Same classifier defect persists across the two archived project states
Guard-to-safeguard binding defect	5 failures / 10 eligible guards	4 failures / 9 eligible guards	Same generator weakness with one changed eligible outcome
Certification-wording template defect	19 / 19 unresolved dossiers	12 / 12 unresolved dossiers	Same unconditional template behavior appears on every unresolved page in each build
Conflict routing mechanism	Same 6 explicit-conflict dossiers, 6 / 6 preserved	Same 6 dossiers, 6 / 6 preserved	Intended routing behavior persists across all shared conflict cases

two NA decisions. A correct control result included flagging a deliberately introduced defect. For example, changing the external query parameter from `query/item-query` to a header or a wrong name triggered the source-role rule. Removing the displayed denominator, disabling the stated zero guard, or replacing the safeguard sentence with a repeated heading triggered the condition rule. Removing a conflict's Review Queue entry or adding certification wording beside unresolved status also triggered the relevant rule. Across the 30 binary controls, the checker missed zero introduced defects and raised zero false alarms; the other two controls returned NA.

4.4. Reviewer Agreement and Checker Results

Across the 112 page–criterion combinations, average pair agreement was 0.634 and Fleiss' κ was 0.446, with one unresolved majority decision. Agreement was 0.553 for condition disclosure, 0.407 for source-role accuracy, 0.452 for disagreement preservation, and −0.285 for review-authority consistency. All 84 authority ratings were explicit Pass or Fail decisions. The negative authority agreement led us to reject that operational definition as a reliable assessment criterion (Fleiss, 1971).

For disagreement preservation, agreement across all 28 cases, including NA judgments, was $\kappa = 0.452$. Restricting the analysis to the seven cases judged applicable by all three reviewers gave $\kappa = -0.167$. This comparison changed which cases were included, rather than treating NA as a missing response. We therefore removed the broad disagreement criterion from the retained assessment dimensions and reported preservation of explicitly marked conflicts as a separate LEAP-specific routing check.

Table 5 reports the full checker comparisons. Figure 4a shows the source-role coverage directly: the checker detected two of the 16 pages identified by reviewer majority and missed the remaining 14, giving 12.5% recall. It also flagged two pages rated Pass by the majority. Two of its four flags were therefore supported, giving 50% precision. For condition disclosure, the comparison across resolved cases gave 40% precision and 100% recall; the three disputed flags are examined below.

Cases C13 and C28 illustrate the source-role coverage limitation. Their source comments describe, respectively, a cross-border transfer of personal data and unsigned trace records, but the published explanations classify those comments as KPI markers. The checker returns Pass for both dossiers, whereas all three reviewers assign source-role Fail. Its implemented patterns detect comments assigned executable roles and SQL cast types mistaken for output aliases; they do not detect these incorrect assignments between documentary roles.

The three condition checker-majority disagreements are C01 (Gross Margin Percentage), C12 (Ethylene Production Yield), and C26 (Yield Percentage). Each has the same vote split: R1=Pass, R2=Pass, R3=Fail, while the checker records Fail. Evidence inspection shows a consistent definition boundary. R1 and R2 use a *source-visibility* criterion: a visible source condition counts as disclosed. R3 and the checker use a *reader-facing* criterion: the published safeguard layer must communicate the condition and bind it to the displayed representation. In C01, the source guard protects `cost == 0` while the displayed business formula denominator is Revenue and the safeguard says only "zero denominator." In C12 and C26, the source visibly contains the zero/non-positive branch while the Logic Safeguards block omits that branch.

Figure 4b shows why condition precision differed: the checker flagged the same five pages, while reviewers differed on whether a condition visible in the source also had to appear in the written explanation. R1 and R2 supplied identical labels on the ten eligible pages and supported two flags; R3 supported all five. The disputed flags were C01, C12, and C26. Precision was therefore 2/5 (40%) against R1/R2 and 5/5 (100%) against R3. The other five eligible pages were Pass for the checker and all reviewers, and recall was 100% under both criteria. The R3 comparison measured agreement with the reader-facing criterion implemented by the checker. Table 7 gives the complete counts.

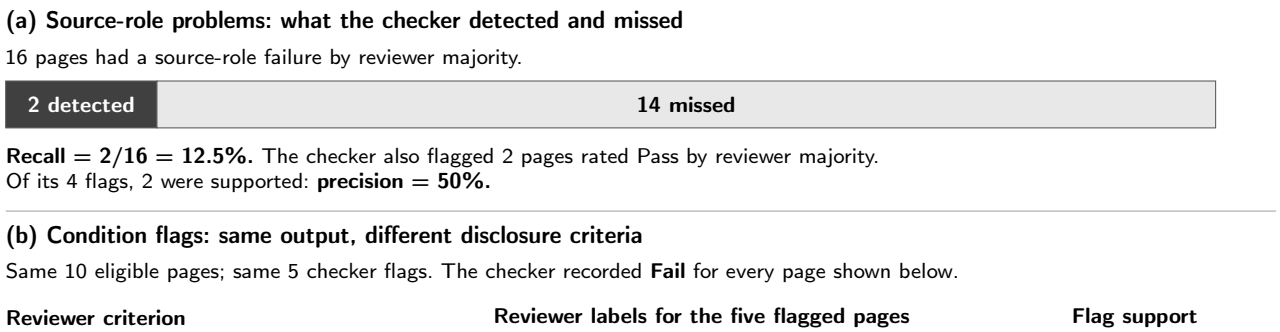


Figure 4: What the automated checks detected and where reviewer judgments differed. (a) The bar partitions the 16 pages with a reviewer-majority source-role failure into two detected and 14 missed pages. The two additional checker flags occurred on majority-Pass pages. (b) The same five condition flags are shown under the two disclosure criteria observed among reviewers on ten eligible pages. R1/R2 supported two flags; R3 supported all five. The R3 row measured agreement with the reader-facing criterion implemented by the checker. Tables 5–7 retain the complete counts and case details.

Table 5
Independent reviewer agreement and unconditional checker comparison. FAIL is the positive class. Precision and recall are computed against reviewer-majority labels on the fixed dossier set.

Candidate	Fleiss' κ	TP	FP	FN	TN	Precision	Recall	Exact / n	Empirical disposition
Source-role accuracy	0.407	2	2	14	10	0.50	0.125	12/28	Dimension retained; released rule fails coverage at 12.5% recall.
Condition disclosure	0.553	2	3	0	22	0.40	1.00	22/27	Dimension retained; majority comparison reflects the source-visibility criterion in three disputed cells.
Disagreement candidate	0.452	0	0	4	24	–	0.00	24/28	Broad criterion removed; agreement on commonly applicable cases was $\kappa = -0.167$; routing reported separately.
Review-authority candidate	−0.285	19	0	3	6	1.00	0.864	19/28	Criterion rejected; all 84 ratings were Pass or Fail, giving $\kappa = -0.285$.

In Phase A, reviewer majority identified issues in 18/28 pages. In Phase B, it identified issues in 23/28. The five additional cases—C03, C08, C11, C20, and C21—all arose from the authority criterion and had the same R1/R3 Fail versus R2 Pass split. These counts describe the two sequential phases; the same reviewers assessed the same cases in that fixed order.

The evaluation retained source-role accuracy and condition disclosure as assessment criteria, reported marked-conflict routing separately, and rejected the review-authority criterion. The source-role rule missed most majority-reference failures. Condition-rule decisions matched the reader-facing labels on all 10 eligible cases and differed from the source-visibility labels on three.

4.5. Comparison with source-linkage verification

We used the same Yield and Gross Margin pages for two checks. First, we checked whether each page identified and exposed the source needed to inspect the calculation. Second, we compared its source-role descriptions and safeguards with that code. We recorded a source-role failure for a wrong

Table 6
Condition cells where checker FAIL conflicts with reviewer-majority PASS. Every cell is a 2–1 split over what counts as disclosure.

Case	Dossier	Majority basis	Checker/R3 basis	Interpretation
C01	Gross Margin Percentage	Visible `cost==0` branch counted as disclosed	Guarded Cost conflicts with displayed Revenue denominator; generic safeguard loses binding	Representation-binding criterion disagreement
C12	Ethylene Production Yield	Visible `feedstock_input_tons <= 0` branch counted as disclosed	Reader-facing safeguards omit the non-positive branch	Source-visibility versus reader-facing-disclosure disagreement
C26	Yield Percentage	Visible `total_input = 0` branch counted as disclosed	Reader-facing safeguards omit the zero-input branch	Source-visibility versus reader-facing-disclosure disagreement

Table 7
Condition-rule sensitivity on the same 10 checker-eligible cases under the two disclosure criteria observed in independent review. FAIL is the positive class. The R3 row measures self-consistency with the reader-facing criterion implemented by the checker; the multi-reviewer analysis provides the independent comparison.

Criterion	TP	FP	FN	TN	Precision	Recall
Source visibility (R1/R2)	2	3	0	5	0.40	1.00
Reader-facing publication (R3)	5	0	0	5	1.00	1.00

description of a source element and a condition failure for an omitted branch or a safeguard that referred to the wrong variable or formula. Table 8 reports the two sets of findings separately.

Both dossiers pass the source-linkage baseline. Yield's fidelity inspection identifies a source-role failure because a function docstring is described as an output alias, and a condition-disclosure failure because its reader-facing safeguards omit the return-zero branch for non-positive feedstock. Gross Margin's condition inspection identifies a representation-binding discrepancy: the safeguard loses the identity of the guarded denominator relative to the displayed formula. Its explicit formula conflict remains preserved in review routing. The condition findings in this comparison use the reader-facing criterion; the source-visibility judgments and their disagreement with that criterion are reported in Table 6. These paired observations show precisely what semantic inspection adds after source availability has been established.

Table 8
Source-linkage baseline and additional explanation-fidelity findings in the two detailed cases.

Case	Source linkage / provenance	Explanation-fidelity result
Ethylene Production Yield	Pass: source statement and executable guard retained	Source-role Fail; condition-disclosure Fail under the reader-facing criterion
Gross Margin Percentage	Pass: declared and executable representations retained with review state	Condition-disclosure Fail under the reader-facing binding criterion; explicit-conflict routing preserved

5. Detailed Case Analysis

5.1. Ethylene Production Yield

Listing 1 shows the Yield source, including its percentage calculation and non-positive-input guard. The dossier displays the positive-input ratio as its Formal Formula. The frozen source uses `feedstock_input_tons <= 0` and returns 0.0 on that branch. The reader-facing Logic Safeguards block omits this non-positive-input behavior. For the docstring shown in Listing 1, the generated lineage prose is "Output alias: maps this expression to a in the result set." Direct source

```
def calculate_ethylene_yield_pct(ethylene_produced_tons: float,
      feedstock_input_tons: float) -> float:
    """Calculates ethylene production yield as a percentage of feedstock
      input."""
    if feedstock_input_tons <= 0:
        return 0.0
    return (ethylene_produced_tons / feedstock_input_tons) * 100.0
```

Listing 1: Verbatim Yield source excerpt from the frozen dossier source; SHA-256 identity is retained in the supplement.

comparison identifies the statement as a function docstring, establishing the lineage-typing defect. The condition finding illustrates the criterion boundary exposed by independent review: two reviewers counted the visible source branch as disclosure, whereas one reviewer and the checker required the reader-facing safeguard layer to state the branch.

5.2. Gross Margin Percentage

Let R be revenue and C be cost. The declared Gross Margin formula, D, divides revenue minus cost by revenue. The implemented expression, I, divides revenue by cost. For illustration, revenue 200 and cost 100 give $D = 50\%$ and $I = 200\%$. The two expressions retained in the dossier are

$$D = 100\frac{R-C}{R}, \qquad I = 100\frac{R}{C}.$$

For $R > 0$ and $C > 0$,

$$\begin{aligned} I - D &= 100\left(\frac{R}{C} + \frac{C}{R} - 1\right) \\ &= 100\left(1 + \frac{(R-C)^2}{RC}\right) \geq 100. \end{aligned}$$

Thus the implemented expression exceeds the declaration by at least 100 percentage points throughout the strictly positive domain, with equality in this lower bound when $R = C$. At positive revenue and zero cost, the declaration evaluates to 100 while the implementation guard returns zero; at zero revenue and positive cost, the declaration is undefined while the implementation returns zero.

The dossier correctly routes the explicit disagreement through *Needs Review* and a Review Queue. Its safeguard text nevertheless describes a generic zero denominator even though the declared and executable representations use different denominators, and certification-oriented language appears alongside unresolved status. The page preserves explicit-conflict routing while also exhibiting a guard-binding discrepancy and certification wording beside an unresolved review state.

6. Discussion

6.1. Interpretation of the Core Findings

The Introduction asked whether a generated KPI page correctly explained the calculation it linked to. The Yield and Gross Margin cases showed why source access and explanation accuracy required separate checks: the code remained available while the page omitted a behavior-changing branch or lost the identity of the guarded denominator. Tracing those errors to the generator answered RQ1 by locating where the explanation went wrong. Recurring errors across archived outputs and alternative inputs were traced to shared generation rules; the changing guard outcomes showed that the same generator could describe some inputs correctly and others incompletely (RQ2).

The reviewer comparison separated agreement on an assessment criterion from the coverage of its automated rule (RQ3–RQ4). Source-role accuracy and condition disclosure received positive reviewer agreement, yet the source-role rule detected only two of 16 majority-reference failures. Agreement on what should be checked and success in detecting it were therefore distinct results. The other candidates required different treatment. Negative agreement led us to reject review-authority consistency. Negative agreement on commonly applicable disagreement cases led us to retain only the narrower marked-conflict routing check. Preserving a marked conflict and accurately explaining a calculation remained separately assessable properties.

6.2. Relation to Prior Research

The observed errors align with code–documentation consistency research. Xu et al. (2024) developed a method for detecting inconsistent code comments that addressed noise and labeling errors in training data. Zhou et al. (2017) compared API documentation with code to detect defects in parameter constraints and exception directives. The LEAP findings connect these concerns to generated KPI pages: the SQL example misdescribed a source element, while Yield omitted an implemented condition. Our study extended the comparison to business-facing formula, explanation, and safeguard fields, then traced errors to the generator. Separately assessing reviewer agreement and checker coverage exposed both ambiguity in the criterion and gaps in the implemented patterns.

Traceability and inconsistency-management research help explain what the retained evidence contributed. Mäder and Gotel (2012) studied maintenance of traceability links as software models changed; our audit examined explanations after the relevant source links were already present. The links made the disputed calculation inspectable. Nuseibeh et al. (2001) described retaining and managing known inconsistencies as part of software development. Gross Margin provided a concrete instance: both formulas and their review routing remained visible, making their disagreement available for inspection. The inaccurate safeguard on that same page showed why preserving a conflict and explaining the associated calculation are complementary responsibilities.

6.3. Why the Checks Disagreed with Reviewers

Figure 3 shows how a generator pattern produced a source-role error: a generic `AS` match selected the SQL conversion type instead of the final output alias. The checker targeted specified patterns of this kind and returned all expected control decisions. Cases C13 and C28 exposed a different gap: documentary comments were called KPI markers, whereas

the checker targeted comments described as executable code and cast types described as aliases. Those cases explain how correct execution of prepared controls coexisted with low recall on reviewer-identified errors (Figure 4a). They identify a missing class of role distinctions for subsequent rule development.

The condition result arose through a different mechanism (Figure 4b). The checker flagged the same five pages under both comparisons. R1/R2 accepted source visibility as disclosure, whereas R3 required the written safeguard to state the condition and its variable. Consequently, the same three flags changed from unsupported to supported when the reference criterion changed, producing precision of 40% and 100%. The difference came from how disclosure was judged. For Yield, the disputed information was present in code but absent from safeguards; for Gross Margin, the safeguard lost the relationship between the guarded variable and the displayed denominator. These cases explain why an explicit disclosure definition is needed to interpret checker performance.

6.4. Practical and Scientific Implications

The worked cases provide concrete review questions. For Yield, a documentation maintainer should check that the explanation states the return-zero behavior for nonpositive feedstock. For Gross Margin, the maintainer should identify the denominator being guarded and preserve both conflicting expressions and the review state. The source excerpt, generated description, and located discrepancy form a reviewable record connecting the business explanation to implementation evidence. Generator tracing then identifies the rule or template to inspect when the same error appears on several pages. This is a practical use of the evidence produced by the audit.

For subsequent checker development, the findings support two separate priorities: extend source-role patterns to the documentary distinctions illustrated by C13/C28, and specify the condition-disclosure criterion before collecting comparison judgments. A subsequent evaluation should test revised rules on additional codebases and retain separate measurements of reviewer agreement, rule conformance, and detection coverage. These measures address different questions about an audit. Comparative documentation studies and cross-role interpretation studies would then address correction effort, business readers' understanding, and joint review with developers as distinct outcomes.

6.5. Study Limitations

The 28 pages came from one author-developed system; shared generation rules produced repeated errors, so page counts describe this output rather than independent generators. The earlier and alternative-input outputs extended the within-system comparison, and the four FastAPI tutorial examples established adapter execution. Broader recurrence requires evidence from other generators. The three-reviewer majority was a comparison reference, with overall $\kappa = 0.446$, rather than a separately adjudicated final set. Reviewer disagreement therefore affects estimates measured against that reference. Original NA labels indicated inapplicability; restricting analysis to common applicability changed the case set and the disagreement estimate. The source-visibility and reader-facing criteria were identified in the returned judgments, so the R3 comparison measures agreement with the criterion implemented by the checker.

The fixed phase order and inclusion of both phase instructions in one packet prevented separation of checklist effects from familiarity or prior rubric access. The increase from 18/28 to 23/28 flagged pages was descriptive; all five additions arose from the rejected authority criterion. Root-cause inspection traced existing output to generator code; a prospective correction-and-retest evaluation remains future work. The routing check covered explicitly marked conflicts, and the condition rule covered comparisons with zero. Near-zero tolerances and function-domain cases require separate rules: real square roots require nonnegative inputs, logarithms positive inputs, and inverse sine/cosine inputs in $[-1, 1]$; inverse tangent accepts all real inputs, and positive-base exponentiation allows negative exponents. The recorded outcomes concern documentation and checking. Human comprehension, correction time, and collaborative benefit require dedicated outcome studies.

7. Conclusion

We audited 28 source-linked KPI documentation pages and found descriptions that assigned source elements the wrong role or omitted conditions that changed the calculation. Comparing checker decisions with three reviewers' judgments showed which errors the implemented rules detected and missed. Inspection of the generator code traced repeated discrepancies to their causes. Reviewer agreement supported source-role accuracy and condition disclosure as the two retained assessment criteria.

The source-role checker achieved 12.5% recall. On the same 10 eligible condition cases, precision was 40% under source visibility and 100% under reader-facing disclosure, with 100% recall under both. The 100% comparison measured agreement with the criterion the checker implemented. The broad disagreement criterion was narrowed to a separate check of explicit-conflict routing, which was preserved in 6/6 cases; negative reviewer agreement led us to reject the review-authority criterion. The initial majority issue set contained 18 of 28 pages.

The contribution is a source-grounded audit and an evaluation of its automated checks. The worked cases show why a reader needs both access to the code and an accurate explanation of it. The evidence identifies the faulty descriptions, their generator causes, and the gaps that remain in the checking rules.

Data and artifact availability

The frozen technical supplement contains the 28 published pages, historical and alternative-input outputs, source and page hashes, audit rules, control labels, scripts, generator excerpts, and the Yield and Gross Margin sources. The unchanged independent-evaluation archive contains the

returned-form derivatives, normalized analysis forms, case mapping, reviewer agreement and checker comparisons, and analysis scripts. Files in that archive's raw directory are intermediate exports: 94 original NA labels appear there as blanks. The accompanying reporting-correction supplement preserves the original obligation responses, a cell-by-cell reconciliation, and both protocol documents. The reconciliation restores the original NA values and matches all 336 labels used in the analysis.

Declaration of competing interest

The authors declare that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this work, the author used Elsevier LeapSpace to support literature discovery, manuscript organization, language refinement, and consistency checking. The authors take responsibility for the content of the article.

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